

1 So Long, and Thanks for All the Seeds: Attacking 2 GGM-trees in Post-quantum signatures

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5 **Abstract. Keywords:** GGM tree · Zero-knowledge signatures · Fault
6 attacks · Post-quantum cryptography · Seed compression

7 1 Introduction

8 Digital signatures form the backbone of several aspects of modern digital life,
9 ranging from firmware updates and secure boot mechanisms to messaging applications
10 and public-key infrastructures. Their primary goal is to guarantee authenticity:
11 in principle, only the holder of the secret signing key should be able to produce
12 a valid signature for a given message. Additionally, digital signatures provide
13 integrity and non-repudiation, ensuring that a message has not been modified
14 and that the signer cannot later deny having produced the signature. Because
15 of their widespread deployment in security-critical systems, the compromise of
16 a digital signature scheme may have severe consequences, including malicious
17 software updates, device impersonation, and unauthorized access to sensitive
18 communications.

19 The security of widely used classical signature schemes, such as RSA-based
20 constructions and elliptic-curve-based signatures relying on the Elliptic Curve
21 Discrete Logarithm Problem, would be compromised by sufficiently large quantum
22 computers through algorithms such as Shor’s algorithm. This threat has motivated
23 the development of post-quantum cryptography, whose goal is to design cryptographic
24 schemes resistant to both classical and quantum adversaries. Among the most
25 promising candidates are zero-knowledge-based signature schemes relying on
26 Fiat–Shamir transformations of interactive identification protocols. Several recent
27 constructions, including LESS [2], CROSS [3], and MEDS [7], use GGM-tree-like
28 seed compression techniques to reduce signature size by compactly representing
29 large collections of ephemeral randomness. During signing, these schemes selectively
30 reveal only the seeds associated with public rounds while keeping challenge-
31 dependent seeds hidden. Although this mechanism is essential for efficiency,
32 recent works have shown that it also introduces an important fault-attack surface:
33 faults affecting the seed-publication logic may leak hidden seeds together with
34 their corresponding zero-knowledge responses, enabling the recovery of secret
35 information and, ultimately, full key recovery or forgery attacks [11,10].

1.1 Related Work

Several recent works have studied fault attacks against post-quantum signature schemes relying on GGM-tree or seed-tree compression techniques, including LESS [2], CROSS [3], MEDS [7], and MPC-in-the-Head-based schemes [6,5].

In [10], the authors introduce fault attacks against the seed-tree compression mechanism used in zero-knowledge-based code-based signatures, primarily targeting LESS-v1 and extending the analysis to CROSS and MEDS. By faulting the seed-publication process, the attacks disclose hidden ephemeral seeds and enable recovery of secret information.

The work of [11] revisits these attacks for LESS-v2, whose incomplete GGM-tree invalidates the original attack strategy. The authors identify new fault-injection points in the incomplete tree-generation and seed-publication procedures, showing that recovering secret-equivalent information is sufficient for universal forgery.

More recently, [9] proposes an alternative attack on LESS-v2 that avoids targeting the tree structure itself. Instead, the attack exploits iteration-skip and loop-abort faults in the preprocessing step that transforms the digest vector before tree generation, enabling high-probability full key recovery.

Our work unifies these attacks under a single Generic ZK Seed Tree (GZKST) abstraction, formalizing the correctness and seed-hiding invariants implicitly exploited by prior works, while additionally providing practical fault attacks against MEDS.

Our Contributions. This work provides a unified analysis of fault attacks against post-quantum signature schemes relying on seed-tree compression mechanisms. Our main contributions are as follows:

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this contributions

- **Generic Seed-Tree Abstraction.** We introduce the *Generic ZK Seed Tree* (GZKST) model, a unified abstraction capturing the seed-tree generation used in schemes such as LESS-v2, MEDS, and CROSS.
- **Unified Analysis of Existing Attacks.** We show that previous fault attacks against LESS-v1 and LESS-v2 can be interpreted as concrete violations of the same seed-hiding invariant, despite targeting different implementation layers and tree constructions.
- **Query Complexity Analysis.** We provide bounds on the number of effective faulted signatures required for complete secret recovery and show that, for all MEDS parameter sets, only a small number of successful queries is sufficient in practice.
- **Practical Fault Injection Attacks on MEDS.** We present practical clock-glitch fault injection attacks against the MEDS reference implementation on an ARM Cortex-M4 microcontroller using a ChipWhisperer-Lite platform. We identify multiple exploitable fault surfaces in the tree traversal logic, including root leakage, hidden-leaf disclosure, and subtree desynchronization attacks.

2 Background

2.1 GGM trees

Pseudorandom functions In cryptography, a core challenge is generating randomness that is computationally indistinguishable from true randomness, yet reproducible given a secret key. A **pseudorandom function** (PRF) addresses this need: it is a keyed function that, to any efficient adversary without knowledge of the key, behaves identically to a truly random function. Intuitively, a PRF stretches a short secret key into an exponentially large lookup table of “random-looking” outputs, without ever materialising that table in memory.

PRFs are foundational primitives in symmetric cryptography. They underpin the construction of message authentication codes (MACs), stream ciphers, and key-derivation functions, among many other schemes. As a concrete example, consider AES used in counter mode: given a secret key k and an index i , the output $\text{AES}_k(i)$ is computationally indistinguishable from a uniformly random string, making it a widely deployed candidate PRF in practice.

A central construction of PRFs from one-way functions is the **Goldreich–Goldwasser–Micali (GGM) tree** [8], which builds a PRF by iteratively applying a pseudorandom generator along the branches of a binary tree indexed by the input bits. GGM trees are the primary PRF construction studied in this work.

Definition 1 (GGM Tree). Let $G : \{0,1\}^\lambda \rightarrow \{0,1\}^{2\lambda}$ be a length-doubling pseudorandom generator, decomposed as $G(x) = (G_0(x), G_1(x))$, where G_0 and G_1 each output λ bits. Given a root seed $\text{seed}_r \in \{0,1\}^\lambda$ and a depth d , the GGM tree is a complete binary tree of depth d in which:

- the root holds seed_r ;
- each internal node v storing seed s generates its left child as $G_0(s)$ and its right child as $G_1(s)$;
- the 2^d leaves are the G outputs at depth d .

The crucial property exploited by ZK signature schemes is the *punctured-PRF property*: given all leaf seeds *except* leaf i , it is computationally infeasible to recover the seed at leaf i , provided G is a secure pseudorandom generator. This makes the GGM tree suitable for hiding one or more seeds while revealing the rest, as shown in Figure 1.

Incomplete GGM trees. When the number of required leaves t is not a power of 2, a scheme may use an *incomplete* tree: leaves appear at multiple levels, and internal nodes near the bottom may have fewer than two children or act as leaves themselves. This reduces memory and computation, but the additional index-tracking logic introduces new fault surfaces, as detailed in Section ??.

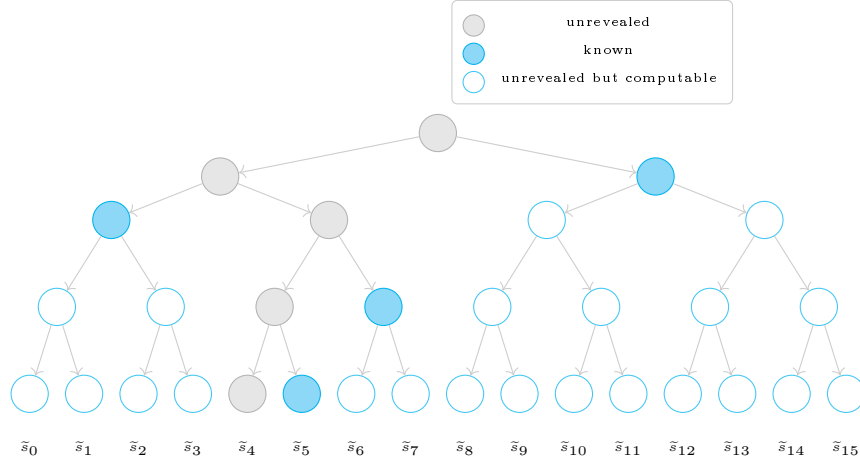


Fig. 1: Receiver’s GGM tree of depth 4. Given only the blue nodes and the PRNG function, the receiver can retrieve all leaves except x_4 .

2.2 Attack targets

This section provides an overview of the signature schemes, to which we apply our generalized model and against which we carry out our attacks.

LESS-v2 LESS (Linear Equivalence Signature Scheme) [2] uses a Fiat–Shamir transformation over a 3-pass Sigma protocol whose security rests on the Canonical Form Linear Equivalence Problem (CF-LEP). During signing, the signer must generate t independent ephemeral monomial matrices $\mathbf{Q}_{e_0}, \dots, \mathbf{Q}_{e_{t-1}} \in \text{Mon}_n(q)$.

Role of the GGM tree in signing and verification. Each leaf $\tilde{s}_i$, for $0 \leq i < t$, is an ephemeral seed of length λ bits from which the i -th monomial map $\mathbf{Q}_{e_i} \in \text{Mon}_n(q)$ is derived by expanding $\tilde{s}_i$ through a PRNG. The tree thus compactly encodes all t independent ephemeral matrices required by the iterated Sigma protocol from a single master seed $\text{mseed} \in \{0, 1\}^\lambda$.

During signing, the Fiat–Shamir challenge vector $\mathbf{b} = (b[0], \dots, b[t-1]) \in \mathcal{S}_{t,w}$ partitions the rounds into two sets: the w rounds with $b[i] \neq 0$, for which the signer publishes $\text{CosetRep}(\pi_{b[i]} \circ \tilde{\pi}_i^*)$ and must *conceal* $\tilde{s}_i$, and the remaining $t-w$ rounds, for which $\tilde{s}_i$ itself is disclosed so that the verifier can re-derive $\mathbf{Q}_{e_i}$ and re-compute the commitment $\text{CF}(A_i)$.

On the verifier side, each received node is expanded downward by the PRNG until all required leaves are reconstructed; the hidden leaves, corresponding to rounds with $b[i] \neq 0$, are never reachable from any published node, so the ephemeral seeds $\tilde{s}_i$ remain protected.

137 *Secret-extraction opportunity.* For every round i with $b[i] \neq 0$, the signer publishes
 138 $\text{CosetRep}(\pi_{b[i]} \circ \tilde{\pi}_i^*)$ while keeping $\tilde{s}_i$ hidden. If an attacker obtains both values
 139 simultaneously, Theorem 2 of [11] shows that the secret permutation $\pi_{b[i]}$ is
 140 recoverable. The flag tree is the *sole enforcement mechanism* that prevents this
 141 dual revelation. Table 1 lists the LESS-v2 parameter sets.

142 CROSS

143 *Context.* CROSS (Codes and Restricted Objects Signature Scheme) [3] is based
 144 on the Restricted Syndrome Decoding Problem (R-SDP) [4] and applies the Fiat–
 145 Shamir transform to a 5-pass (q_2 -identification) protocol. Like MEDS, CROSS
 146 uses a *binary* fixed-weight challenge vector $\mathbf{c} = (c_0, \dots, c_{t-1}) \in \{0, 1\}^t$ of weight
 147 w , so the hidden and revealed index sets are $\mathcal{H} = \{i : c_i = 1\}$ and $\mathcal{R} = \{i : c_i =$
 148 $0\}$, identical in structure to MEDS.

149 *Role of the seed tree in signing and verification.* Each leaf Seed_i , for $0 \leq i < t$,
 150 is a λ -bit ephemeral seed from which the pair of randomness vectors $(\mathbf{e}'_i, \mathbf{u}'_i) \in$
 151 $G \times \mathbb{F}_p^n$ is derived via $\text{CSPRNG}(\text{Seed}_i)$. These vectors are the sole secret material
 152 for round i : from them the signer computes the linear transitive map $\mathbf{v}_i =$
 153 $\mathbf{e} \star (\mathbf{e}'_i)^{-1} \in G$ and the two commitments $\text{cmt}_0[i] = \text{Hash}((\mathbf{v}_i \star \mathbf{u}'_i) \mathbf{H}^\top \mid \mathbf{v}_i)$ and
 154 $\text{cmt}_1[i] = \text{Hash}(\mathbf{u}'_i \mid \mathbf{e}'_i)$. The seed tree thus compactly encodes all t pairs of
 155 ephemeral vectors required by the iterated Sigma protocol from a single master
 156 seed.

157 During signing, the Fiat–Shamir second challenge vector $\text{chall}_2 \in \mathcal{B}(t, w)$ of
 158 fixed Hamming weight w partitions the rounds into two sets: the w rounds with
 159 $\text{chall}_2[i] = 1$, for which the signer publishes the explicit response $\text{resp}[i] = (\mathbf{y}_i, \mathbf{v}_i)$
 160 and must *conceal* Seed_i , and the remaining $t - w$ rounds with $\text{chall}_2[i] = 0$,
 161 for which Seed_i itself is disclosed so that the verifier can re-expand $(\mathbf{e}'_i, \mathbf{u}'_i)$,
 162 recompute $\mathbf{y}_i = \mathbf{u}'_i + \text{chall}_1[i] \cdot \mathbf{e}'_i$, and recover $\text{cmt}_1[i]$.

163 Additionally, CROSS uses a parallel *Merkle tree* over the t commitments $\text{cmt}_0[i]$
 164 to avoid transmitting all t digests explicitly: for the w rounds where $\text{chall}_2[i] = 1$,
 165 the corresponding $\text{cmt}_0[i]$ values cannot be recomputed by the verifier (since
 166 Seed_i is hidden), and are instead authenticated via a Merkle proof of size $O(w \log t)$.

167 On the verifier side, the received seed-path nodes are expanded downward to
 168 reconstruct all unrevealed leaf seeds; the w hidden seeds, corresponding to rounds
 169 with $\text{chall}_2[i] = 1$, are provably unreachable from any published node, ensuring
 170 that the ephemeral vectors $(\mathbf{e}'_i, \mathbf{u}'_i)$ — and hence the secret key $\mathbf{e}$ — remain
 171 protected.

172 *Secret-extraction opportunity.* For round i with $c_i = 1$, the signer reveals a ZK
 173 response encoding the relationship between $\tilde{s}_i$ and the long-term secret R-SDP
 174 solution. If an attacker obtains both the response and $\tilde{s}_i$, the secret is recoverable
 175 by the linear-algebra argument of ZKFault [10], which demonstrated full key
 176 recovery from a single fault for all CROSS parameter sets (see Table 1).

MEDS MEDS (Matrix Equivalence Digital Signature) [7] is a code-based signature scheme whose security rests on the Matrix Code Equivalence Problem (MCEP). Like LESS, it applies the Fiat–Shamir transform to a 3-pass Sigma protocol between a prover holding a secret equivalence $(A, B) \in \text{GL}_k(q) \times \text{GL}_n(q)$ and a verifier.

Role of the seed tree in signing and verification. Each leaf σ_i , for $0 \leq i < t$, is an ephemeral seed of length $\ell_{\text{tree_seed}}$ bytes from which the i -th pair of ephemeral invertible matrices $(\tilde{A}_i, \tilde{B}_i) \in \text{GL}_m(q) \times \text{GL}_n(q)$ is sampled via a PRNG. The commitment for round i is then $h_i = H(\text{SF}(\pi_{\tilde{A}_i, \tilde{B}_i}(G_0)))$, so the seed tree compactly encodes all t ephemeral matrix pairs required by the iterated Sigma protocol from a single master seed $\rho_{0,0} \in \mathcal{B}^{\ell_{\text{tree_seed}}}$.

During signing, the Fiat–Shamir challenge string $\mathbf{h} = (h_0, \dots, h_{t-1}) \in \{0, \dots, s-1\}^t$ of fixed weight w partitions the rounds into two sets: the w rounds with $h_i \neq 0$, for which the signer publishes the coset representative $(\mu_i, \nu_i) = (A\tilde{A}_i^{-1}, B^{-1}\tilde{B}_i)$ and must *conceal* σ_i , and the remaining $t-w$ rounds with $h_i = 0$, for which σ_i is disclosed so that the verifier can re-expand $(\tilde{A}_i, \tilde{B}_i)$ and recover the commitment h_i .

On the verifier side, the required leaf seeds are recovered; the w hidden seeds, corresponding to rounds with $h_i \neq 0$, are provably unreachable from any published node.

Since we are going to show in §4 an application of the attack in the reference implementation of MEDS, we are going to explain better the algorithms of key generation, signing and verification. Table 1 shows the parameters for MEDS [7].

Key Generation. Algorithm 1 is the simplification of the key generation. It begins by sampling a master secret seed δ and immediately deriving two sub-seeds via XOF: a *public* seed σ_{G_0} , used to expand the base systematic matrix $G_0 \in \mathbb{F}_q^{k \times mn}$, and a *private* chaining seed σ that is used in the rest of the process. For each of the $s-1$ public keys, the algorithm performs a *compressed key-generation* step: a secret invertible matrix $T_i \in \text{GL}_k(q)$ twists G_0 into $G'_0 = T_i G_0$, and a structured linear system (the Solve step, made efficient by fixing $P_0 = I_m$ and $P_1 = U_m$) recovers the equivalence pair $(A_i, B_i) \in \text{GL}_m(q) \times \text{GL}_n(q)$ satisfying $G_i = \text{SF}(\pi_{A_i, B_i}(G_0))$. The public key stores only the seed σ_{G_0} and the compressed matrices $G_1, \dots, G_{s-1}$; the secret key retains δ together with the stored inverses A_i^{-1}, B_i^{-1} —the sole secret material consumed at signing time.

Signing. Algorithm 2 is the simplification of the signing algorithm. A fresh master seed δ is drawn and expanded into a tree-root seed ρ and a salt α ; the seed tree $\text{SeedTree}_t(\rho, \alpha)$ then materialises all t leaf seeds $\sigma_0, \dots, \sigma_{t-1}$ at once. For every round i , a pair of ephemeral matrices $(\tilde{A}_i, \tilde{B}_i) \in \text{GL}_m(q) \times \text{GL}_n(q)$ is expanded from σ_i (together with α and an address for domain separation), and the commitment $\tilde{G}_i = \text{SF}(\pi_{\tilde{A}_i, \tilde{B}_i}(G_0))$ is computed. Hashing all non-systematic parts of the $\tilde{G}_i$ alongside the message yields the digest d , which is parsed into the

Table 1: Cryptographic parameters for LESS-v2 [2], MEDS [7], and CROSS [3]
 (*: MEDS uses $n=m=k$; CROSS uses $q=2$ implicitly via $\mathbb{F}_p$).

Scheme	Parameter set	t	w	s	Sec.	Extra
LESS-v2	LESS-252-45	45	34	8	I	$n=252, k=126, q=127$
	LESS-252-68	68	42	4	I	$n=252, k=126, q=127$
	LESS-252-192	192	36	2	I	$n=252, k=126, q=127$
	LESS-400-102	102	61	4	III	$n=400, k=200, q=127$
	LESS-400-220	220	68	2	III	$n=400, k=200, q=127$
	LESS-548-137	137	79	4	V	$n=548, k=274, q=127$
	LESS-548-345	345	75	2	V	$n=548, k=274, q=127$
MEDS*	MEDS-9923	1152	14	4	I	$n=14, q=4093$
	MEDS-13220	192	20	5	I	$n=14, q=4093$
	MEDS-41711	608	26	4	III	$n=22, q=4093$
	MEDS-69497	160	36	5	III	$n=22, q=4093$
	MEDS-134180	192	52	5	V	$n=30, q=2039$
	MEDS-167717	112	66	6	V	$n=30, q=2039$
CROSS	CROSS-R-SDP-1	163	60	2	I	$p=127, n=68, k=8$
	CROSS-R-SDP-3	200	75	2	III	$p=127, n=80, k=9$
	CROSS-R-SDP-5	250	94	2	V	$p=127, n=90, k=9$
	CROSS-R-SDPe-1	55	25	2	I	$p=509, n=60, k=6$
	CROSS-R-SDPe-3	75	34	2	III	$p=509, n=80, k=8$
	CROSS-R-SDPe-5	90	40	2	V	$p=509, n=90, k=8$

218 fixed-weight challenge vector $(h_0, \dots, h_{t-1})$. For each *non-zero* challenge h_i , the
 219 signer releases the coset representative $(\mu_i, \nu_i) = (\tilde{A}_i A_{h_i}^{-1}, B_{h_i}^{-1} \tilde{B}_i)$; the remaining
 220 seeds are transmitted compactly via a seed-tree path p .

221 *Verification.* Algorithm 3 is the simplification of the verification algorithm. The
 222 verifier first reconstructs the public matrices $G_0, G_1, \dots, G_{s-1}$ from the public
 223 key and parses the signed message to recover the path p , the digest d , the salt
 224 α , and the w response pairs. The challenge vector $(h_0, \dots, h_{t-1})$ is recomputed
 225 from d , and **PathToSeedTree** _{t} recovers the $t - w$ disclosed leaf seeds from p .
 226 For each round i , one of two paths is taken: if $h_i > 0$, the verifier applies
 227 the published coset representative (μ_i, ν_i) to the corresponding public matrix
 228 G_{h_i} and checks invertibility before computing $\hat{G}_i = \text{SF}(\pi_{\mu_i, \nu_i}(G_{h_i}))$; if $h_i = 0$,
 229 the verifier re-expands $(\hat{A}_i, \hat{B}_i)$ from the recovered seed and computes $\hat{G}_i =$
 230 $\text{SF}(\pi_{\hat{A}_i, \hat{B}_i}(G_0))$. Finally, the recomputed digest d' is compared to d ; equality
 231 confirms the signature.

232 *Secret-extraction opportunity.* For round i with $h_i = 1$, the signer reveals its ZK
 233 response. If an attacker obtains both the response and $\tilde{s}_i$, the secret equivalence
 234 pair (A, B) is recoverable by linear algebra. The Reference Tree is the sole
 235 enforcement mechanism preventing this dual revelation.

Algorithm 1: MEDS Key Generation (simplified)

Input: —
Output: Public key pk, secret key sk

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1  $\delta \xleftarrow{\$} \mathcal{B}^{\ell_{\text{sec\_seed}}}$ 
2  $\sigma_{G_0}, \sigma \leftarrow \text{XOF}(\delta, \ell_{\text{pub\_seed}}, \ell_{\text{sec\_seed}})$ 
3  $G_0 \leftarrow \text{ExpandSystMat}(\sigma_{G_0})$ 
4 for  $i = 1$  to  $s - 1$  do
5    $\sigma_a, \sigma_{T_i}, \sigma \leftarrow \text{XOF}(\sigma, \ell_{\text{sec\_seed}}, \ell_{\text{sec\_seed}}, \ell_{\text{sec\_seed}})$ 
6    $T_i \leftarrow \text{ExpandInvMat}(\sigma_{T_i}, k)$ 
7    $a_{m-1, m-1} \leftarrow \text{ExpandFqs}(\sigma_a, 1)$ 
8    $G'_0 \leftarrow T_i G_0$ 
9    $(A_i, B_i^{-1}) \leftarrow \text{Solve}(G'_0, a_{m-1, m-1})$  // Fail  $\Rightarrow$  retry from line 5
10   $G_i \leftarrow \text{SF}(\pi_{A_i, B_i}(G_0))$  //  $\perp \Rightarrow$  retry from line 5
11  $\text{pk} \leftarrow (\sigma_{G_0} | \text{CompressG}(G_1) | \dots | \text{CompressG}(G_{s-1}))$ 
12  $\text{sk} \leftarrow (\delta | \sigma_{G_0} | \text{Compress}(A_1^{-1}) | \dots | \text{Compress}(A_{s-1}^{-1}) | \text{Compress}(B_1^{-1}) | \dots | \text{Compress}(B_{s-1}^{-1}))$ 
13 return pk, sk

```

236 3 Abstract Model

237 We now define a unified abstraction that captures the common structure of the
238 GGM-tree mechanism across all schemes discussed in Section 2.

239 **Definition 2 (GZKST).** A Generic ZK Seed Tree (GZKST) is a tuple $\Pi =$
240 (Setup, Expand, Chal, Decide, Publish, Recon) parametrised by:

- 241** – λ : the security parameter;
- 242** – $t \in \mathbb{N}$: the number of parallel ZK repetitions;
- 243** – $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda}$: a secure pseudorandom generator;
- 244** – $\mathcal{S}$: the secret type (e.g., permutations for LESS, matrix pairs for MEDS,
245 party shares for MPCitH).

246 The six algorithms are:

- 247** 1. Setup(1^λ) $\rightarrow$ (seed_r, salt): samples a master seed seed_r $\xleftarrow{\$} \{0, 1\}^\lambda$ and a
248 public salt.
- 249** 2. Expand(seed_r, salt) $\rightarrow$ ($\mathcal{T}, (\tilde{s}_0, \dots, \tilde{s}_{t-1})$): builds a binary tree $\mathcal{T}$ by top-down
250 G-expansion from seed_r. Each leaf $i \in \{0, \dots, t-1\}$ holds a seed $\tilde{s}_i$. Tree
251 topology may be complete or incomplete (scheme-dependent).
- 252** 3. Chal(cmt, msg) $\rightarrow$ $\mathbf{c} = (c_0, \dots, c_{t-1})$: derives a Fiat-Shamir challenge from
253 the commitment and message. $c_i = 0$ indicates that round i requires the
254 ephemeral seed; $c_i \neq 0$ indicates a secret-dependent response.
- 255** 4. Decide($\mathbf{c}$) $\rightarrow$ ($\mathcal{H}, \mathcal{R}$): partitions $\{0, \dots, t-1\}$ into the hidden index set $\mathcal{H} =$
256 $\{i : c_i \neq 0\}$ and the revealed index set $\mathcal{R} = \{i : c_i = 0\}$.
- 257** 5. Publish($\mathcal{T}, \mathcal{H}$) $\rightarrow$ path: constructs a flag structure $\mathcal{F} : V(\mathcal{T}) \rightarrow \{0, 1\}$ and
258 returns the minimal set of nodes path from which every $\tilde{s}_i$ with $i \in \mathcal{R}$ is
259 reconstructible, while no $\tilde{s}_i$ with $i \in \mathcal{H}$ is derivable.

Algorithm 2: MEDS Signing (simplified)

Input: Secret key sk , message m
Output: Signed message m_s

- 1 Parse sk to recover σ_{G_0} , and A_i^{-1} , B_i^{-1} for $i = 1, \dots, s-1$
- 2 $G_0 \leftarrow \text{ExpandSystMat}(\sigma_{G_0})$
- 3 $\delta \xleftarrow{\$} \mathcal{B}^{\ell_{\text{sec_seed}}}$
- 4 $\rho, \alpha \leftarrow \text{XOF}(\delta, \ell_{\text{tree_seed}}, \ell_{\text{salt}})$
- 5 $(\sigma_0, \dots, \sigma_{t-1}) \leftarrow \text{SeedTree}_t(\rho, \alpha)$
- 6 **for** $i = 0$ **to** $t-1$ **do**
- 7 $\sigma'_i \leftarrow (\alpha | \sigma_i | \text{ToBytes}(2^{1+\lceil \log_2 t \rceil} + i, 4))$
- 8 $\sigma_{\tilde{A}_i}, \sigma_{\tilde{B}_i} \leftarrow \text{XOF}(\sigma'_i, \ell_{\text{pub_seed}}, \ell_{\text{pub_seed}})$
- 9 $\tilde{A}_i \leftarrow \text{ExpandInvMat}(\sigma_{\tilde{A}_i}, m); \quad \tilde{B}_i \leftarrow \text{ExpandInvMat}(\sigma_{\tilde{B}_i}, n)$
- 10 $\tilde{G}_i \leftarrow \text{SF}(\pi_{\tilde{A}_i, \tilde{B}_i}(G_0)) \quad // \perp \Rightarrow \text{resample } \sigma'_i$
- 11 $d \leftarrow H(\text{Compress}(\tilde{G}_0[; k, mn-1]) | \dots | \text{Compress}(\tilde{G}_{t-1}[; k, mn-1]) | m)$
- 12 $(h_0, \dots, h_{t-1}) \leftarrow \text{ParseHash}_{s,t,w}(d)$
- 13 **for** $i = 0$ **to** $t-1$ **with** $h_i > 0$ **do**
- 14 $\mu_i \leftarrow \tilde{A}_i \cdot A_{h_i}^{-1}; \quad \nu_i \leftarrow B_{h_i}^{-1} \cdot \tilde{B}_i$
- 15 Append $(\text{Compress}(\mu_i) | \text{Compress}(\nu_i))$ to response vector v
- 16 $p \leftarrow \text{SeedTreeToPath}_t(h_0, \dots, h_{t-1}, \rho, \alpha)$
- 17 **return** $m_s \leftarrow (v | p | d | \alpha | m)$

260 6. $\text{Recon}(\text{path}, c, \text{salt}) \rightarrow \{\tilde{s}_i : i \in \mathcal{R}\}$: verifier-side reconstruction of revealed
 261 leaf seeds from path.

262 **3.1 The Flag Structure**

263 The flag structure $\mathcal{F}$ produced internally by **Publish** is the key intermediate
 264 object. It is a labelling $\mathcal{F} : V(\mathcal{T}) \rightarrow \{0, 1\}$ where:

- 265 – $\mathcal{F}(v) = 0$: node v is *publishable*—either directly included in **path** or derivable
 266 from a published ancestor;
- 267 – $\mathcal{F}(v) = 1$: node v is *hidden*—it must not appear in **path** and must not be
 268 derivable from published nodes.

269 The flag structure satisfies two invariants:

270 **Invariant 1 (Correctness).** For every $i \in \mathcal{R}$ there exists a node v on the path
 271 from the root to leaf $\tilde{s}_i$ such that $\mathcal{F}(v) = 0$ and $\mathcal{F}(\text{parent}(v)) = 1$.

272 **Invariant 2 (ZK hiding).** For every $i \in \mathcal{H}$, no ancestor of leaf $\tilde{s}_i$ satisfies
 273 $\mathcal{F}(v) = 0$.

274 The ZK property of the signature scheme is contingent on Invariant 2 holding for
 275 all $i \in \mathcal{H}$. A fault that changes $\mathcal{F}(v)$ from 1 to 0 for any v that is an ancestor of
 276 some $\tilde{s}_i$ with $i \in \mathcal{H}$ breaks Invariant 2 and may allow an attacker to reconstruct
 277 $\tilde{s}_i$.

Algorithm 3: MEDS Verification (simplified)

Input: Public key pk , signed message m_s
Output: Message m or $\perp$

```

1  $G_0 \leftarrow \text{ExpandSystMat}(\text{pk}[0, \ell_{\text{pub\_seed}} - 1])$ 
2 for  $i = 1$  to  $s - 1$  do
3    $G_i \leftarrow \text{DecompressG}(\text{pk}[\dots])$ 
4 Parse  $m_s$  to obtain: response pairs  $((\mu_i, \nu_i))_{h_i > 0}$ , path  $p$ , digest  $d$ , salt  $\alpha$ , message  $m$ 
5  $(h_0, \dots, h_{t-1}) \leftarrow \text{ParseHash}_{s,t,w}(d)$ 
6  $(\sigma_0, \dots, \sigma_{t-1}) \leftarrow \text{PathToSeedTree}_t(h_0, \dots, h_{t-1}, p, \alpha)$ 
7 for  $i = 0$  to  $t - 1$  do
8   if  $h_i > 0$ 
9      $(\mu_i, \nu_i) \leftarrow$  next response pair from  $m_s$ 
10    if  $\mu_i \notin \text{GL}_m(q)$  or  $\nu_i \notin \text{GL}_n(q)$  return  $\perp$ 
11     $\hat{G}_i \leftarrow \text{SF}(\pi_{\mu_i, \nu_i}(G_{h_i}))$ 
12    if  $\hat{G}_i = \perp$  return  $\perp$ 
13  else
14     $\sigma'_i \leftarrow (\alpha | \sigma_i | \text{ToBytes}(2^{1+\lceil \log_2 t \rceil} + i, 4))$ 
15     $\hat{A}_i \leftarrow \text{ExpandInvMat}(\text{XOF}(\sigma'_i)_1, m)$ ;  $\hat{B}_i \leftarrow \text{ExpandInvMat}(\text{XOF}(\sigma'_i)_2, n)$ 
16     $\hat{G}_i \leftarrow \text{SF}(\pi_{\hat{A}_i, \hat{B}_i}(G_0))$ 
17  $d' \leftarrow H(\text{Compress}(\hat{G}_0[k, mn-1]) \dots \text{Compress}(\hat{G}_{t-1}[k, mn-1]) | m)$ 
18 if  $d = d'$  return  $m$ 
19 return  $\perp$ 

```

3.2 Mapping Schemes to the GZKST Model

We now establish that each of the ZK-based post-quantum signature schemes introduced in Section 2 is a valid GZKST instance in the sense of Definition 2. For each scheme, we exhibit the concrete instantiation of the six algorithms and verify that Invariants 1 and 2 hold by construction.

LESS-V2

Proposition 1 (LESS-v2 is a GZKST instance). *LESS-v2 instantiates Definition 2 with secret type $\mathcal{S} = \{\pi_1, \dots, \pi_{s-1}\}$ (the $s - 1$ secret monomial permutations), t parallel repetitions, and an incomplete GGM tree of height $\lceil \log_2 t \rceil$. Invariants 1 and 2 hold by construction of the three-phase flag tree; see Appendix A.1 for the full proof.*

Remark 1 (Invariants under incomplete trees). When the tree is incomplete, some leaves appear at depth $\lceil \log_2 t \rceil - 1$. The index-tracking arrays `npl`, `ll`, `off`, and `lsi` ensure that `LabelLeaves` and `ComputeSeeds` visit the correct nodes regardless of level. Both invariants therefore hold for any tree shape produced by the LESS-v2 expansion (see Appendix A.1).

- `Setup( $1^\lambda$ )` $\rightarrow$ (`seedr`, `salt`): The root seed is derived from the secret key seed; the salt is sampled uniformly at random.
- `Expand(seedr, salt)` $\rightarrow$ ($\mathcal{T}, (\tilde{s}_0, \dots, \tilde{s}_{t-1})$): LESS-v2 uses an incomplete GGM tree with $\lceil \log_2 t \rceil$ levels. Leaves are distributed across the bottom two levels; index offsets are tracked via `npl`, `ll`, `off`, and `lsi` (Figure 2a).

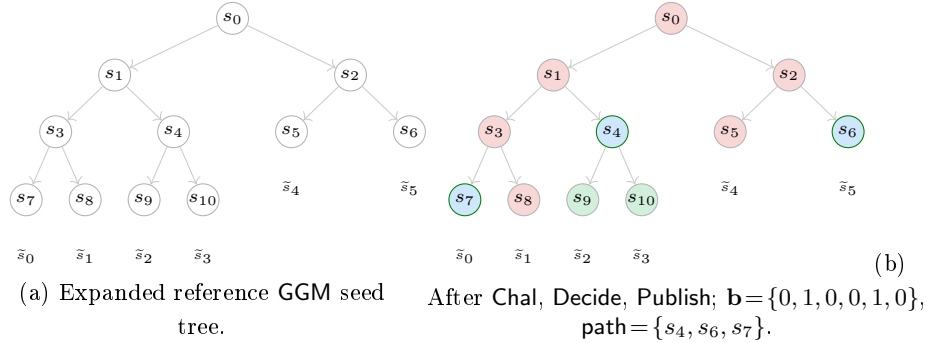


Fig. 2: LESS-v2 GGM seed tree ($t = 6$): structure (a) and flag state (b). In (b): pink nodes ($\mathcal{F}=1$, hidden subtree); green nodes ($\mathcal{F}=0$, fully revealed subtree); blue-bordered nodes ($\mathcal{F}=0$, in path).

- 299 – $\text{Chal}(\text{cmt}, \text{msg}) \rightarrow \mathbf{c}$: The t canonical-form matrices together with the salt
- 300 and message are hashed; **SampleChallenge** produces $\mathbf{b} \in \{0, \dots, s-1\}^t$ of
- 301 weight w . Setting $c_i = b^{(i)}$, the condition $c_i = 0$ (resp. $c_i \neq 0$) signals a
- 302 revealed (resp. hidden) round.
- 303 – $\text{Decide}(\mathbf{c}) \rightarrow (\mathcal{H}, \mathcal{R})$: $\mathcal{H} = \text{supp}(\mathbf{b})$, $\mathcal{R} = \{0, \dots, t-1\} \setminus \mathcal{H}$.
- 304 – $\text{Publish}(\mathcal{T}, \mathcal{H}) \rightarrow \text{path}$: LESS-v2 builds an auxiliary flag tree of the same
- 305 shape via two phases: (i) *leaf labelling* (**LabelLeaves**): $\mathcal{F}(\text{leaf}_i) = \mathbf{1}[i \in \mathcal{H}]$;
- 306 (ii) *upward propagation* (**ComputeSeeds**): $\mathcal{F}(v) = \mathcal{F}(\ell) \vee \mathcal{F}(r)$ for each internal
- 307 node v with children ℓ and r . The two-phase description is an implementation
- 308 detail; the resulting flag assignment is exactly the OR-propagation of equations (1)–
- 309 (2) (the root’s flag is determined solely by this propagation). The path
- 310 $\text{path} = \{v : \mathcal{F}(v) = 0, \mathcal{F}(\text{parent}(v)) = 1\}$ (with the convention that
- 311 $\mathcal{F}(\text{parent}(\text{root})) = 1$) is the minimal publishable boundary (Figure 2b).
- 312 – $\text{Recon}(\text{path}, \mathbf{c}, \text{salt}) \rightarrow \{\tilde{s}_i : i \in \mathcal{R}\}$: **PathToSeedTree** places received nodes
- 313 on an empty tree of the same shape and re-expands downward via **G** until
- 314 all t leaves are filled. Positions $i \in \mathcal{H}$ remain empty; positions $i \in \mathcal{R}$ yield
- 315 the recovered seeds from which monomial matrices and commitments are
- 316 re-derived.

317 MEDS

318 **Proposition 2 (MEDS is a GZKST instance).** *MEDS instantiates Definition 2*
 319 *with secret type $\mathcal{S} = \{(A_j^{-1}, B_j^{-1})\}_{j=1}^{s-1}$, t parallel repetitions, and a GGM tree*
 320 *of height $\lceil \log_2 t \rceil$ (complete when t is a power of two, incomplete otherwise;*
 321 *only the first t of the $2^{\lceil \log_2 t \rceil}$ leaves are used). Invariants 1 and 2 hold by*
 322 *the SeedTreeToPath construction regardless of whether the tree is complete or*
 323 *incomplete; see Appendix A.2.*

- 324 – $\text{Setup}(1^\lambda) \rightarrow (\text{seed}_r, \text{salt})$: MEDS samples a fresh random scalar and derives
- 325 the tree seed and salt via a single XOF call.
- 326 – $\text{Expand}(\text{seed}_r, \text{salt}) \rightarrow (\mathcal{T}, (\tilde{s}_0, \dots, \tilde{s}_{t-1}))$: A binary tree of height $\lceil \log_2 t \rceil$ is
- 327 built by hashing each parent with the salt and node index; the first t leaves
- 328 are returned. When t is not a power of two, the rightmost $2^{\lceil \log_2 t \rceil} - t$ potential
- 329 leaves are unused, making the tree incomplete—the same structural situation
- 330 as LESS-v2.
- 331 – $\text{Chal}(\text{cmt}, \text{msg}) \rightarrow \mathbf{c}$: Each leaf is expanded into sub-seeds via XOF, ephemeral
- 332 matrices are sampled, and the commitment is the hash of all compressed
- 333 generator matrices together with m . Parsing the digest via $\text{ParseHash}_{s,t,w}$
- 334 yields $\mathbf{h} \in \{0, \dots, s-1\}^t$ of weight w ; setting $c_i = h_i$, $c_i = 0$ (resp. $c_i \neq 0$)
- 335 signals a revealed (resp. hidden) round.
- 336 – $\text{Decide}(\mathbf{c}) \rightarrow (\mathcal{H}, \mathcal{R})$: $\mathcal{H} = \{i : c_i \neq 0\}$, $\mathcal{R} = \{i : c_i = 0\}$.
- 337 – $\text{Publish}(\mathcal{T}, \mathcal{H}) \rightarrow \text{path}$: SeedTreeToPath_t removes the label of each leaf $i \in \mathcal{H}$
- 338 and propagates each deletion upward, erasing every ancestor that has at least
- 339 one hidden leaf in its subtree (i.e., the same bottom-up $\mathcal{F}(v) = \mathcal{F}(\ell) \vee \mathcal{F}(r)$
- 340 rule as LESS-v2). Surviving boundary nodes—those with $\mathcal{F}(v) = 0$ and
- 341 $\mathcal{F}(\text{parent}(v)) = 1$ —are serialised in depth-first order.
- 342 – $\text{Recon}(\text{path}, \mathbf{c}, \text{salt}) \rightarrow \{\tilde{s}_i : i \in \mathcal{R}\}$: PathToSeedTree_t places path nodes on an
- 343 empty tree and re-expands downward via G; positions $i \in \mathcal{H}$ remain empty.

344 *Remark 2.* In this instantiation of the Publish algorithm, we rely on the path
 345 construction as described in the MEDS submission document [?]. However, the
 346 reference implementation uses a different algorithm, that is further explained in
 347 section 4.

348 CROSS

349 **Proposition 3 (CROSS is a GZKST instance).** *CROSS instantiates Definition 2*
 350 *with secret type $\mathcal{S} = \{\mathbf{e}\}$ (the R-SDP solution), t parallel repetitions, and a*
 351 *complete GGM tree. The seed-publishing component satisfies Invariants 1 and 2*
 352 *by the same argument as MEDS (Proposition 2). The additional Merkle proof*
 353 *over $\{\text{cmt}_0[i]\}_{i \in \mathcal{H}}$ is a CROSS-specific commitment-authenticity mechanism orthogonal*
 354 *to the GZKST seed-hiding invariants.*

- 355 – $\text{Setup}(1^\lambda) \rightarrow (\text{seed}_r, \text{salt})$: Root seed of size λ bits and salt of size 2λ bits,
- 356 both drawn uniformly at random.
- 357 – $\text{Expand}(\text{seed}_r, \text{salt}) \rightarrow (\mathcal{T}, (\tilde{s}_0, \dots, \tilde{s}_{t-1}))$: A complete binary tree (CROSS-
- 358 v1 identical to LESS-v1; CROSS-v2 uses the SeedTreePath mechanism of
- 359 MEDS) where each parent seed is concatenated with the salt and node index
- 360 and processed by SHAKE to yield two λ -bit child seeds.
- 361 – $\text{Chal}(\text{cmt}, \text{msg}) \rightarrow \mathbf{c}$: CROSS applies Fiat–Shamir in two rounds. The t
- 362 commitment pairs $(\text{cmt}_0[i], \text{cmt}_1[i])$, the salt, and the message are hashed
- 363 to produce $\text{chall}_1 \in (\mathbb{F}_p^*)^t$, from which first-response digests $y_i = \text{Hash}(\mathbf{u}'_i +$
- 364 $\text{chall}_1[i] \cdot \mathbf{e}'_i)$ are derived. A second hash yields $\text{chall}_2 \in \mathcal{B}(t, w)$; setting

365 $c_i = \text{chall}_2[i] \in \{0, 1\}$, $c_i = 0$ (resp. 1) signals a revealed (resp. hidden)
 366 round.
 367 – $\text{Decide}(\mathbf{c}) \rightarrow (\mathcal{H}, \mathcal{R})$: $\mathcal{H} = \{i : c_i = 1\}$, $\mathcal{R} = \{i : c_i = 0\}$.
 368 – $\text{Publish}(\mathcal{T}, \mathcal{H}) \rightarrow \text{path}$: The seed path is computed as in MEDS. Additionally,
 369 a Merkle proof over $\{\text{cmt}_0[i]\}_{i \in \mathcal{H}}$ authenticates the w commitment values
 370 that the verifier cannot recompute; this is orthogonal to the GZKST seed-
 371 hiding invariants (see Proposition 3).
 372 – $\text{Recon}(\text{path}, \mathbf{c}, \text{salt}) \rightarrow \{\tilde{s}_i : i \in \mathcal{R}\}$: Path nodes are placed on an empty tree
 373 and re-expanded downward. Positions $i \in \mathcal{H}$ remain empty; positions $i \in \mathcal{R}$
 374 yield seeds from which $(\mathbf{e}'_i, \mathbf{u}'_i)$ are re-derived and $\text{cmt}_1[i]$ is recomputed for
 375 verification.

376 **Definition 3 (Faulted Signing Oracle).** *Fix a node $v^* \in V(\mathcal{T})$. The faulted*
 377 *signing oracle $\mathcal{O}^{v^*}(\text{sk}, \text{pk})$ takes a message msg and executes the signing algorithm*
 378 *with a single modification: during $\text{Publish}(\mathcal{T}, \mathcal{H})$, the flag value is forced to*

$$\mathcal{F}(v^*) \leftarrow 0,$$

379 *regardless of whether $\mathcal{F}(v^*)$ equals 1 in the honest execution. Upward propagation*
 380 *is applied as normal after this assignment. The oracle returns the resulting*
 381 *(possibly faulted) signature sig^* .*

382 3.3 Oracle Abstraction

383 We previously presented the abstract view of the GGM-tree and its instantiations
 384 in LESS, MEDS, and CROSS. We now formalize the attacker's interface in an
 385 abstract manner, extending the GZKSTuple with a fault model and a secret-
 386 extraction procedure.

387 *Faulted signing oracle.* Let $\Pi = (\text{Setup}, \text{Expand}, \text{Chal}, \text{Decide}, \text{Publish}, \text{Recon})$ be
 388 a GZKST instance with secret type $\mathcal{S}$, and let (sk, pk) be a fixed key pair. We
 389 model the adversary's access to the victim's signing device as a *faulted oracle*:

390 **Definition 4 (Faulted Signing Oracle).** *Fix a node $v^* \in V(\mathcal{T})$. The faulted*
 391 *signing oracle $\mathcal{O}^{v^*}(\text{sk}, \text{pk})$ takes a message msg and executes the signing algorithm*
 392 *with a single modification: during $\text{Publish}(\mathcal{T}, \mathcal{H})$, the flag value is forced to*

$$\mathcal{F}(v^*) \leftarrow 0,$$

393 *regardless of whether $\mathcal{F}(v^*)$ equals 1 in the honest execution. Upward propagation*
 394 *is applied as normal after this assignment. The oracle returns the resulting*
 395 *(possibly faulted) signature sig^* .*

396 The fault is *effective* if the original value $\mathcal{F}(v^*)$ was 1 and v^* is an ancestor of at
 397 least one leaf $\tilde{s}_i$ with $i \in \mathcal{H}$: in this case, the published path path now contains
 398 a node from which $\tilde{s}_i$ is derivable, directly violating Invariant 2. The fault is
 399 *ineffective* otherwise (the node was already publishable, or all hidden leaves lie
 400 in the sibling subtree).

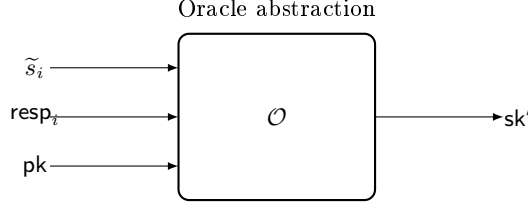


Fig. 3: The **Extract** oracle: given dual revelation $(\tilde{s}_i, \text{resp}_i)$ and public key pk , it outputs partial secret sk' .

Remark 3. The fault model of Definition 4 is realised by several concrete physical mechanisms: an instruction-skip fault that bypasses the store $\mathcal{F}(v) \leftarrow \mathcal{F}(\ell) \vee \mathcal{F}(r)$, a stuck-at-zero perturbation forcing $\mathcal{F}(v^*) = 0$, a bit-flip $(1 \rightarrow 0)$ via Rowhammer, or a clock-glitch that skips the validity check on a specific tree node. All of these are documented in the literature [10,11].

Remark 4. We assume that our faulted oracle has an algorithm **SolverS** that efficiently solves secrets of type $\mathcal{S}$. For example, when instantiating our oracle against LESS, we use Lemma 1 from [10] to solve the secret type.

Dual revelation. The invariant that the **Publish** phase must enforce is that, for every hidden round $i \in \mathcal{H}$, the signer simultaneously publishes:

- (i) the ZK *response* resp_i , which encodes the relationship between $\tilde{s}_i$ and the long-term secret sk ; and
- (ii) a path path from which $\tilde{s}_i$ is *not* derivable.

An effective fault breaks condition (ii): the attacker now holds both $\tilde{s}_i$ (reachable from the faulted path) and resp_i (always present in the signature). This *dual revelation* is the primitive from which all secret-extraction procedures in this work are built.

Secret extraction. We extend the GZKST tuple with a seventh, scheme-specific procedure:

Definition 5 (Extract). $\text{Extract}(\tilde{s}_i, \text{resp}_i, \text{pk}) \rightarrow \text{sk}'$: given the leaked seed $\tilde{s}_i$ and the ZK response resp_i for the same round $i \in \mathcal{H}$, recover the (partial) secret $\text{sk}' \subseteq \mathcal{S}$.

For the schemes studied in this work, **Extract** takes the following concrete forms:

- **MEDS.** For a hidden round i with $h_i = j$, the signer publishes $(\mu_i, \nu_i) = (A_j \tilde{A}_i^{-1}, B_j^{-1} \tilde{B}_i)$ (consistent with the background description in §2.2). Expanding $\tilde{s}_i$ yields $(\tilde{A}_i, \tilde{B}_i)$, so

$$A_j = \mu_i \cdot \tilde{A}_i, \quad B_j^{-1} = \nu_i \cdot \tilde{B}_i^{-1}.$$

Algorithm 4: GZKST key recovery from faulted oracle

Input : Faulted oracle $\mathcal{O}^{v^*}$, public key pk
Output : Recovered secret key $\hat{\text{sk}}$

```

1 Collected  $\leftarrow \emptyset$ 
2 while  $|\text{Collected}| < |\mathcal{S}|$  do
3    $\text{sig}^* \leftarrow \mathcal{O}^{v^*}(\text{msg})$  for an arbitrary msg
4   if  $\text{sig}^*$  is effective
5     for each  $i \in \text{leaked}(\text{sig}^*)$  do
6        $\text{sk}' \leftarrow \text{Extract}(\tilde{s}_i, \text{resp}_i, \text{pk})$ 
7        $\text{Collected} \leftarrow \text{Collected} \cup \{(j, \text{sk}')\}$  where  $j = c_i$ 
8 return  $\hat{\text{sk}} \leftarrow \text{Collect}$ 

```

427 Both operations are direct matrix multiplications (and one inversion) over
428 $\mathbb{F}_q$; no column-reconstruction step is required.
429 – **LESS-v2**. The response is a coset representative $\text{CosetRep}(\pi_{b[i]} \circ \tilde{\pi}_i^*)$. Expanding
430 $\tilde{s}_i$ gives the partial monomial matrix $\tilde{\pi}_i^*$, and the secret permutation $\pi_{b[i]}$ is
431 recovered via the column-extraction argument of Lemma 1 in [11].
432 – **CROSS**. The response encodes $\mathbf{v}_i = \mathbf{e} \star (\mathbf{e}'_i)^{-1}$. Expanding $\tilde{s}_i$ gives $\mathbf{e}'_i$, so
433 $\mathbf{e} = \mathbf{v}_i \star \mathbf{e}'_i$ directly.

434 *Full key recovery.* When $|\mathcal{S}| = s - 1 > 1$ (as in MEDS with $s \geq 3$, and LESS
435 with $s \geq 3$), a single effective fault recovers only the secret indexed by h_i . Full
436 recovery requires observing each index $j \in \{1, \dots, s - 1\}$ at least once.

437 **Proposition 4 (Correctness of Algorithm 4).** *Let Π be a GZKST instance,*
438 *suppose $v^* = \text{root}$ (so the fault exposes all w hidden leaves, making $L = w$*
439 *deterministic), and let $p_{\min} = 1 - \left(\frac{s-2}{s-1}\right)^w$. Algorithm 4 terminates and outputs*
440 *$\hat{\text{sk}} = \text{sk}$ with probability at least $1 - \delta$ after at most $\left\lceil \frac{s-1}{p_{\min}} \cdot \ln \frac{1}{\delta} \right\rceil$ effective oracle*
441 *queries; see Appendix A.4 for the full proof.*

442 *Remark 5 (General fault location).* For an effective fault at an arbitrary node v^*
443 with ℓ leaf descendants, the number $L = |\mathcal{H} \cap \text{leaves}(v^*)|$ of hidden leaves in the
444 subtree is a hypergeometric random variable whose realisation may be smaller
445 than its mean $\bar{w} = w\ell/t$, making the per-query success probability $p_r(L) =$
446 $1 - \left((s-1-r)/(s-1)\right)^L$ potentially smaller than the root-fault $p_{\min}$. A valid bound
447 for any effective fault is obtained by taking the worst case $L = 1$, which gives
448 $p_r \geq r/(s-1)$ for all $r \geq 1$ and therefore $\mathbb{E}[Q] \leq (s-1)H_{s-1}$, where $H_{s-1} =$
449 $\sum_{r=1}^{s-1} 1/r$ is the $(s-1)$ -th harmonic number. For $s \leq 6$ (all schemes in Table 1),
450 this gives $\mathbb{E}[Q] \leq 5H_5 \approx 11.4$, still a small constant.

451 Detecting whether a received signature is effective is feasible for the adversary:
452 given the fault location v^* and the challenge $\mathbf{c}$ (recoverable from the digest d in

the signature), the adversary recomputes the honest flag tree, checks whether $\mathcal{F}(v^*) = 1$, and verifies that at least one hidden leaf falls in the subtree of v^* . This detection is detailed for LESS in [10] and applies unchanged to MEDS and CROSS.

Expected query count. Let ℓ denote the number of leaves in the subtree rooted at v^* .

Proposition 5. *Let $v^* = \text{root}$, so $L = w$ exactly. Define $p_{\min} = 1 - \left(\frac{s-2}{s-1}\right)^w$. The per-query success probability satisfies $p_r \geq p_{\min}$ for all $r \geq 1$, and the expected total number of effective queries satisfies*

$$\mathbb{E}[Q] \leq \frac{s-1}{p_{\min}}.$$

See Appendix A.5 for the derivation.

Remark 6 (Role of E_{dist} and Jensen’s inequality). The quantity $E_{\text{dist}} = (s-1)p_{\min}$ equals the expected number of distinct secret indices revealed per effective query (when $L = w$ is deterministic at the root). When $v^* \neq \text{root}$ and L is random with mean $\bar{w} = w\ell/t$, Jensen’s inequality (the function α^x is convex in x) gives $\mathbb{E}[\#\text{distinct}] \leq E_{\text{dist}}$: E_{dist} is an *upper* bound on expected per-query gain, which implies a *lower* bound on $\mathbb{E}[Q]$, not an upper bound. The upper bound $\mathbb{E}[Q] \leq (s-1)/p_{\min}$ stated in Proposition 5 follows from the direct inequality $p_r \geq p_{\min}$, not from the Jensen argument.

For all MEDS parameter sets (Table 1), faulting the root ($v^* = \text{root}$, $\ell = t$) gives $\bar{w} = w$; for those parameters $w \gg 1$, so $p_{\min} \approx 1$ and $\mathbb{E}[Q] \lesssim s-1$, which is at most 5 across all sets.

Remark 7 (Concrete query counts for MEDS). Table 2 reports $p_{\min}$ and $\mathbb{E}[Q]$ for every MEDS parameter set when $v^* = \text{root}$ (so $L = w$ exactly). The values use the exact formula $p_{\min} = 1 - ((s-2)/(s-1))^w$; for all parameter sets $p_{\min} > 0.99$, giving $\mathbb{E}[Q] \leq s-1 \leq 5$.

Table 2: Expected effective queries $\mathbb{E}[Q]$ for MEDS (fault at root).

Parameter set	s	w	$s-1$	$p_{\min}$	$\mathbb{E}[Q] \leq$
MEDS-9923	4	14	3	0.9966	3.01
MEDS-13220	5	20	4	0.9968	4.01
MEDS-41711	4	26	3	1.0000	3.00
MEDS-69497	5	36	4	1.0000	4.00
MEDS-134180	5	52	4	1.0000	4.00
MEDS-167717	6	66	5	1.0000	5.00

478 *From key recovery to forgery.* Once Algorithm 4 terminates, the adversary holds
 479 a complete signing key $\hat{sk}$ and can produce valid signatures on any message
 480 msg^* — including messages never submitted to any oracle. This constitutes a
 481 full break of EUF-CMA:

482 **Corollary 1.** *Let Π be a GZKST-based signature scheme and let the fault model*
 483 *be as in Definition 4 (a stuck-at-zero perturbation of a single flag bit during*
 484 *Publish). If an efficient adversary can induce this fault at a fixed node $v^* \in$*
 485 *$V(\mathcal{T})$ on the victim’s signing device, then Π is not EUF-CMA-secure in the*
 486 *fault model: the adversary recovers sk via Algorithm 4 and forges signatures on*
 487 *arbitrary messages without further oracle access.*

488 4 Fault Injection attacks on MEDS GGM tree procedures

489 This subsection details a fault injection (FI) attack conducted on the C reference
 490 implementation of the MEDS signature algorithm. We focus on exploiting the
 491 tree traversal logic used in the path construction to leak hidden seeds.

492 4.1 Target Analysis

493 The primary target of our analysis is the `SeedTreePath` function, specifically
 494 targeting the logic represented in the signing phase of the MEDS algorithm
 495 (instruction 16 of algorithm 2). In the reference implementation, both `SeedTreePath`
 496 (path generation) and `SeedTree` (tree reconstruction) are handled by a unified
 497 function, `stree_to_path_to_stree`, which operates in two modes defined by a
 498 `mode` parameter.

Listing 1.1: `stree_to_path_to_stree` implementation.

```

499 1 void stree_to_path_to_stree(uint8_t *stree, uint8_t *h_digest, uint8_t *path, uint8_t *salt
500 2     , int mode) {
501 3     int h = 0, i = 0;
502 4     unsigned int id = 0, idx = 0;
503 5     unsigned int indices[MEDS_w] = {0};
504 6     for (int i = 0; i < MEDS_t; i++) {
505 7         if (h_digest[i] > 0) {
506 8             indices[idx++] = i;
507 9         }
508 10    }
509 11    while (1==1) {
510 12        int index_leaf = 0;
511 13        while ((indices[id] > (MEDS_seed_tree_height-h)) == i)
512 14            { // Traverse down toward the secret leaf
513 15                h += 1;
514 16                i <<= 1;
515 17                if (h > MEDS_seed_tree_height) {
516 18                    h -= 1;
517 19                    i >>= 1;
518 20                    if (id + 1 < MEDS_w) id++;
519 21                    index_leaf = 1; // Flag node as hidden
520 22                    break;
521 23                }
522 24            }
523 25    }

```

```

524 24     if (index_leaf == 0) {
525 25         if (mode == STREE_TO_PATH) {
526 26             memcpy(path, &stree[MEDS_st_seed_bytes * SEED_TREE_ADDR(h, i)],
527             MEDS_st_seed_bytes);
528 27             path += MEDS_st_seed_bytes;
529 28         } else {
530 29             memcpy(&stree[MEDS_st_seed_bytes * SEED_TREE_ADDR(h, i)], path,
531             MEDS_st_seed_bytes);
532 30             path += MEDS_st_seed_bytes;
533 31             t_hash(stree, salt, h, i);
534 32         }
535 33     }
536 34     // Backtracking logic
537 35     while ((i & 1) == 1) {
538 36         h -= 1;
539 37         i >>= 1;
540 38     }
541 39     i += 1;
542 40     if ((i << (MEDS_seed_tree_height - h)) >= MEDS_t) return;
543 41 }
544 42 }

```

546 The algorithm identifies secret leaves (where $h[i] > 0$) and performs a Depth-
547 First Search (DFS). The exploration halts as soon as it reaches a node that is
548 no longer an ancestor of the hidden leaf. This node is then appended to the
549 signature path. If the exploration reaches a target hidden leaf, the `index_leaf`
550 flag is set to ensure the leaf seed remains unpublished and we move on to the
551 next hidden leaf.

552 **Objective** The adversary’s goal is to manipulate the control flow via fault
553 injection to leak unpublished seeds (internal nodes or leaves) into the public
554 signature path, ultimately compromising the security of the signature scheme.

555 4.2 Experimental Setup

556 Our experimental platform consists of a **CW303-STM32F3** target board, featuring
557 an ARM Cortex-M4 STM32F303RCT7 processor. Clock glitching is performed
558 using a **ChipWhisperer-Lite**. As a proof-of-concept, the target board is flashed
559 exclusively with the `stree_to_path_to_stree` function, using fixed values for
560 the reference tree, salt, and challenge.

561 4.3 Fault Models

562 We propose three fault models targeting different stages of the tree traversal.

563 **Model 1: Full Tree Recovery (Root Leakage)** This attack aims to reveal
564 the root seed, which effectively compromises the entire tree. By injecting a fault
565 that causes the skip of the first iteration of `while loop` at line 13 of listing 1.1,
566 the DFS halts immediately at the root ($h = 0, i = 0$). The root is incorrectly
567 identified as a node ready for publication.

568 *Detection and recovery.* The signature path contains only a single non-zero
 569 element. Verification is performed by expanding this single seed into a full tree
 570 and checking if the resulting signature is accepted.

571 **Model 2: Subtree Recovery (Index Corruption)** By glitching the loop
 572 `if (id + 1 < MEDS_w) id++;` at line 19 of listing 1.1, the algorithm fails
 573 to update the target index `id`. The traversal logic becomes desynchronized,
 574 resulting in the algorithm treating subsequent branches as public. All leaves
 575 to the right of the faulted `id` become computable.

576 *Detection and recovery.* This fault is hard to detect without a reference signature.
 577 However, one can simulate tree reconstructions for all possible faulted `id` values
 578 to verify which leads to a valid signature. (up to w verifications)

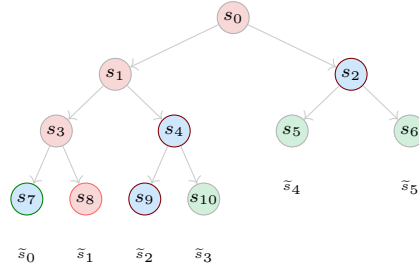


Fig. 4: Effect of fault model 2 on the example presented in figure 2, if the glitch occurred while `id=1`. The returned path is $\{s_7, s_9, s_4, s_2\}$ and all leaves are computable except x_1 .

579 **Model 3: Hidden Leaf Recovery** This model targets the specific instruction
 580 `index_leaf = 1;` at line 20 of listing 1.1. By skipping this assignment, a hidden
 581 leaf is no longer marked as private. The execution continues normally, but the
 582 secret leaf seed is copied into the `path` buffer.

583 *Detection and recovery.* The path length will be $w + k$ (where $k > 0$ is the
 584 number of iterations of the target instruction that were skipped). Detection
 585 involves testing subsets of the path to find a valid signature, which may be
 586 computationally expensive ($\binom{w+k}{w}$ verifications).

587 4.4 Results and Observations

588 We successfully executed the fault models on the Chipwhisperer board. This
 589 section detailed the compiler and glitcher configurations that were required to
 590 obtain the desired faults.

Methodology For each target fault model, we aim to identify the parameter configurations that yield the highest attack success rate. First, we define a target output state that signifies a successful fault injection.

Next, we establish a precise faulting window encompassing one or more target assembly instructions. To synchronize the injection hardware, we instrument the firmware by inserting a `trigger_high()` instruction immediately before the target instruction block to signal the ChipWhisperer to begin counting clock cycles, followed by a `trigger_low()` instruction to terminate the window. We then verify the disassembled binary to ensure that compiler optimizations have not rearranged, omitted, or reordered these trigger placements.

To configure the temporal bounds of the injection, we define the `repeat` parameter, which specifies the continuous duration (in clock cycles) of the clock glitch. We calibrate this maximum value based on the nominal cycle execution counts provided in the ARM documentation for the target instruction sequence.

Finally, we iteratively sweep the following spatial and temporal parameters to map the physical fault space and isolate high-yield vulnerability regions:

- `width`: The width of the glitch pulse, achieved by XORing the system clock (expressed as a percentage of the clock period, spanning from 0% to 50%).
- `offset`: The phase shift or delay introduced before the rising edge of the glitch pulse relative to the clock cycle (expressed as a percentage of the clock period, spanning from -50% to 50%).
- `ext_offset`: The absolute number of clock cycles elapsed after the trigger signal before the glitch injection begins.

Full tree recovery This attack is successful if the returned path contains a single node which is the root node.

Compilation. This attack is carried out on a code compiled with the default `-Os` optimization.

Trigger position. We directly target the evaluation of the conditional and the branching instruction corresponding to line 19 of 1.1). The compiled result is in 1.2. We fault for 3 clock cycles.

Listing 1.2: Compiled trigger points for fault model 1.

```

0800062c <stree_to_path_to_stree_glitch>:
...
// trigger_high();
8000710: f001 fa12 b1 8001b38 <trigger_high>
// while ((indices[id] >> (MEDS_seed_tree_height - h)) == i)
8000714: ab06 add r3, sp, #24
8000716: eb03 0488 add.w r4, r3, r8, lsl #2
800071a: e00d b.n 8000738 <stree_to_path_to_stree_glitch+0x66>
// trigger_low();
800071c: f001 fa13 b1 8001b46 <trigger_low>
...

```

634 *Parameters.* Figure 5 shows that the attack has a high success rate and lower
 635 reset rates for an `ext_offset` ranging from 1.5 to 2.5, a small `offset` of `-1` to
 636 `1` and rather wide glitches (`width` over 20).

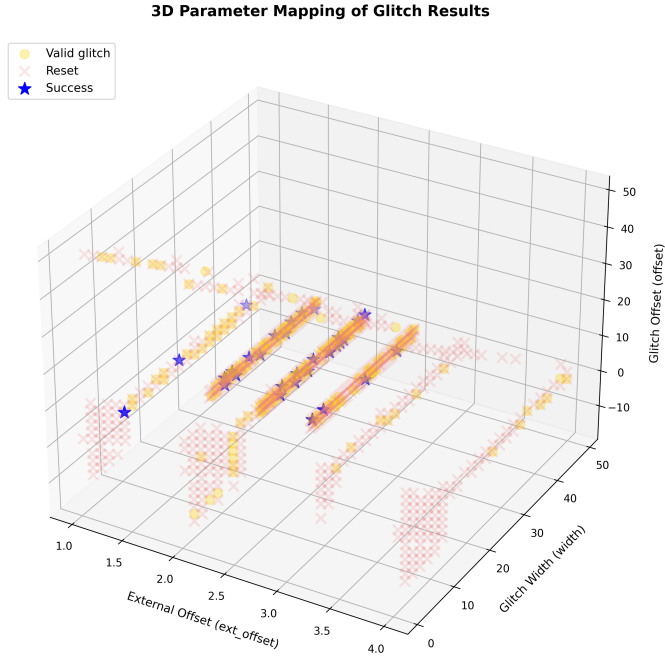


Fig. 5: Glitch success frequency by glitch width and offset for fault model 1, with 10 tries per parameter set. Valid glitches are runs that did not crash and output paths that were neither correct nor the target faulted path.

637 Subtree recovery

638 *Compilation.* This is carried out on a code compiled with the default `-Os`
 639 optimization.

640 *Trigger position.* We set the trigger to frame lines 19 of 1.1. The compiled result
 641 is in 1.3.

Listing 1.3: Compiled trigger points for fault model 2.

```
642 080006d2 <stree_to_path_to_stree_glitch>:
643 ...
644 8000726: f001 fa09 b1 8001b3c <trigger_high> // trigger_high();
645 800072a: f108 0401 add.w r4, r8, #1
646 800072e: 2c05 cmp r4, #5 //if (id+1 < MEDS_w)
```

```

648: 8000730: bf88      it  hi
649: 8000732: 4644      movhi r4, r8      // id++ ;
650: 8000734: f001 fa09  bl  8001b4a <trigger_low> // trigger_low();
651: ...

```

653 *Parameters.* The evaluation of the condition and the incrementation of `id`
654 require 3 clock cycles. We therefore set `repeat` to 3. We consider the attack
655 successful if we obtain the faulted path pattern described by figure 4.

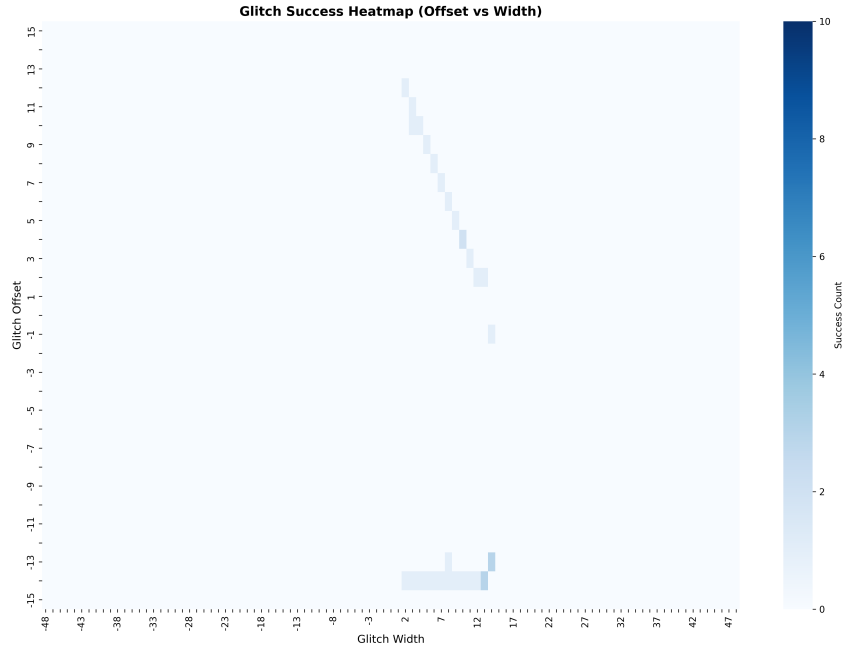


Fig. 6: Glitch success frequency by glitch width and offset for fault model 2, with 10 tries per parameter pair.

656 Single hidden leaf recovery

657 *Compilation.* Higher compiler optimization levels made it difficult to precisely
658 target the flags. Specifically, the `-O1`, `-O2`, `-O3`, and `-Os` optimization flags
659 resulted in compiled assembly code that omitted the explicit assignment instruction
660 entirely, relying instead on complex control-flow manipulations to achieve the
661 same logical outcome. To resolve this issue, we enforced the `-O0` optimization
662 level strictly on the targeted function. Applying `-O0` to the entire project was
663 unfeasible, as it increased the binary size beyond the storage capacity of the
664 target board.

665 *Trigger position.* For the unoptimized version, we position the glitch high trigger
 666 right before the `index_leaf` assignment and the low trigger right after. This
 667 results in the compiled code in listing 1.4.

Listing 1.4: Compiled trigger points for fault model 3.

```

668 0800062c <stree_to_path_to_stree_glitch>:
669 ...
670 80006be: f001 fd3f bl 8002140 <trigger_high> // trigger_high();
671 80006c2: 2301 movs r3, #1 // index_leaf = 1;
672 80006c4: 62bb str r3, [r7, #40] @ 0x28
673 80006c6: f001 fd43 bl 8002150 <trigger_low> // trigger_low();
674 ...
675

```

677 *Parameters.* The disassembled code shows that the assignment is done in two
 678 instructions, first The CPU loads the immediate integer value 1 directly into
 679 the register `r3`, then it takes that value 1 from `r3` and writes it out to memory.
 680 According to ARM documentation, each of these instruction is 1 clock cycle
 681 long. We test the attack by setting the `repeat` parameter of the glitch controller
 682 to 2 and 3.

683 In both cases, we notice a higher probability of success with an offset of 1 and a
 684 positive width, or a width of -1 and a positive offset.

685 **Discussion /conclusion** todo.

686 5 Countermeasures

687 In this section we suggest a set of countermeasures that would make the signature
 688 scheme more secure against the exposed fault models.

689 We split the countermeasures into different categories :

- 690 – Post-publication code path check : same signature size, no modification of
- 691 the tree construction.
- 692 – Path construction algorithm change : same signature size.
- 693 – GGM tree-less signature : larger signature.

694 5.1 Post-publication path check

695 This countermeasure consists is running checks on the output of `SeedTreeToPath`.

696 A simple size check on the path is enough to be secure against fault models 1 and
 697 3, since they result in path lengths of 1 and golden path size+1 respectively. The
 698 expected path size can be computed in linear time from $\mathcal{H}$. (Todo : algorithm
 699 ?).

700 A more thorough and correct check would be to rebuild the tree using `PathToSeedTree`
 701 on the obtained path and to compare the obtained leaves to $\mathcal{H}$ and the reference
 702 tree. This can also be done in linear time.

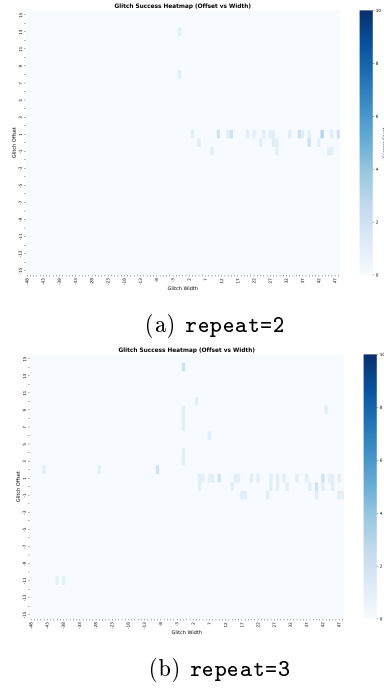


Fig. 7: Glitch success frequency by glitch width and offset for fault model 3, with 10 tries per parameter pair.

Remark 8. In order to pass this check, it would be sufficient to induce the same fault as in the `SeedTreeToPath` on `PathToSeedTree`, seeing as they both call the same function in the reference implementation.

Impact on performances We implement both of these checks and compare execution times of the signature step with or without countermeasure. The results are summarized in table 3. We refer to the path size check countermeasure by C_{size} and the full tree check by C_{tree} .

Table 3: Countermeasures execution time comparison

Version	Time (min)	cycles
Base	0.001230	4798061
C_{size}	0.001070	4171382
C_{tree}	0.001303	5080762

5.2 Path construction algorithm modification

Though the current implementation of `SeedTreeToPath/PathToSeedTree` is very efficient, it allows for attack namely because it merges the flag tree construction and the path construction steps.

Splitting the two steps and introducing a check after both stages could limit the attack surface by introducing redundancy.

Todo : detail + implementation + time analysis.

5.3 GGM-tree-less signature

We could also consider not using GGM trees and publishing all $t-w$ seeds. Todo : size of the signature.

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A Full Proofs

Throughout this appendix, $\mathcal{T}$ denotes a GGM seed tree with leaf set $\{0, \dots, t-1\}$ and $\mathcal{F} : V(\mathcal{T}) \rightarrow \{0, 1\}$ its flag labelling, as produced by the Publish phase of

the relevant GZKST instance. We use the convention $\mathcal{F}(v) = 1$ to mean that v 's subtree contains at least one hidden leaf ($i \in \mathcal{H}$), and $\mathcal{F}(v) = 0$ to mean that all leaf descendants of v are revealed ($i \in \mathcal{R}$).

A.1 Proof of Proposition 1 (LESS-v2 is a GZKST instance)

Let $\mathcal{T}$ be the LESS-v2 seed tree with $\lceil \log_2 t \rceil$ levels. After the three-phase flag-tree construction the flag values are:

$$\mathcal{F}(\text{leaf}_i) = \mathbf{1}[i \in \mathcal{H}], \quad i \in \{0, \dots, t-1\}, \quad (1)$$

$$\mathcal{F}(v) = \mathcal{F}(\ell) \vee \mathcal{F}(r), \quad \text{for every internal node } v \text{ with children } \ell, r. \quad (2)$$

Equations (1)–(2) jointly imply:

$$\mathcal{F}(v) = 1 \iff \exists i \in \mathcal{H} : \text{leaf}_i \text{ is a descendant of } v. \quad (3)$$

The published path is $\text{path} = \{v : \mathcal{F}(v) = 0, \mathcal{F}(\text{parent}(v)) = 1\}$ (with the convention that $\mathcal{F}(\text{parent}(\text{root})) = 1$).

Proof (Proof of Invariant 1 (Correctness)). Let $i \in \mathcal{R}$. Consider the root-to-leaf path $\pi_i = (v_0 = \text{root}, v_1, \dots, v_d = \text{leaf}_i)$.

Since $|\mathcal{H}| \geq 1$ (the challenge weight satisfies $w \geq 1$), at least one leaf has flag 1. By (3), $\mathcal{F}(v_0) = \mathcal{F}(\text{root}) = 1$.

Since $i \in \mathcal{R}$, equation (1) gives $\mathcal{F}(v_d) = 0$.

Because $\mathcal{F}(v_0) = 1$ and $\mathcal{F}(v_d) = 0$, the path π_i contains at least two nodes with different flags. Define $k = \min\{j \in \{0, \dots, d\} : \mathcal{F}(v_j) = 0\}$. Then $k \geq 1$ (since $\mathcal{F}(v_0) = 1$), $\mathcal{F}(v_k) = 0$, and $\mathcal{F}(v_{k-1}) = 1$. Hence $v_k \in \text{path}$, and v_k lies on the root-to-leaf path π_i , establishing Invariant 1.

Proof (Proof of Invariant 2 (ZK hiding)). Let $i \in \mathcal{H}$ and let v be any proper ancestor of leaf_i (including the root). Since leaf_i is a descendant of v and $i \in \mathcal{H}$, equation (3) gives $\mathcal{F}(v) = 1$. Hence no ancestor of leaf_i can satisfy $\mathcal{F}(v) = 0$, so no ancestor of leaf_i lies in path . Invariant 2 holds.

Proof (Extension to incomplete trees (Remark 1)). When t is not a power of 2, the LESS-v2 tree is incomplete: some leaves appear at depth $\lceil \log_2 t \rceil - 1$ rather than $\lceil \log_2 t \rceil$. The index-tracking arrays `npl`, `ll`, `off`, and `lsi` ensure that `LabelLeaves` correctly identifies which nodes are leaves and assigns flags according to (1), and that `ComputeSeeds` performs the bottom-up OR pass (2) over the actual tree structure.

The proofs of both invariants above are purely graph-theoretic and depend only on the properties (1)–(3), which hold regardless of whether leaves appear at a uniform depth. Hence both invariants hold for any incomplete tree shape produced by the LESS-v2 expansion.

810 A.2 Proof of Proposition 2 (MEDS is a GZKST instance)

811 `SeedTreeToPath` implements the following erasure: for each $i \in \mathcal{H}$ the algorithm
 812 (i) removes the label of leaf_i , and (ii) propagates the removal upward—a node
 813 v loses its label as soon as *at least one* leaf in its subtree has been removed.
 814 By induction this is equivalent to the OR-propagation $\mathcal{F}(v) = \mathcal{F}(\ell) \vee \mathcal{F}(r)$ of
 815 LESS-v2.

816 *Claim.* After the erasure pass, a node v retains its label if and only if $\mathcal{F}(v) = 0$
 817 in the sense of (3), i.e., *every* leaf descendant of v is in $\mathcal{R}$ (no hidden leaf in v 's
 818 subtree).

819 *Proof.* ($\Rightarrow$) Suppose v is retained. By definition of the upward propagation, no
 820 ancestor of any hidden leaf is retained. Hence v 's subtree contains no hidden
 821 leaf, i.e., $\mathcal{F}(v) = 0$.

822 ($\Leftarrow$) Suppose $\mathcal{F}(v) = 0$, i.e., every leaf in v 's subtree is in $\mathcal{R}$. The erasure step
 823 removes only hidden leaves and their ancestors. Since v 's subtree contains no
 824 hidden leaf, no removal is triggered in v 's subtree, so v retains its label.

825 The published path `path` consists of the highest surviving nodes in the tree,
 826 serialised in depth-first order. Equivalently,

$$\text{path} = \{v : v \text{ retains its label and } \text{parent}(v) \text{ was erased}\}.$$

827 By the claim, this is $\{v : \mathcal{F}(v) = 0, \mathcal{F}(\text{parent}(v)) = 1\}$, which is identical to the
 828 LESS-v2 definition.

829 Invariants 1 and 2 follow by the same argument as in Appendix A.1.

830 A.3 Proof of Proposition 3 (CROSS is a GZKST instance)

831 The seed-tree component of CROSS (`SeedTreePath`) is structurally equivalent
 832 to MEDS's `SeedTreeToPath` (see Section 3.2). The argument of Appendix A.2
 833 therefore applies directly, establishing Invariants 1 and 2 for the seed-hiding part.

834 The additional Merkle proof over $\{\text{cmt}_0[i]\}_{i \in \mathcal{H}}$ is an authenticity mechanism for
 835 the commitment digests that the verifier cannot recompute from the revealed
 836 seeds. It neither publishes any seed in $\{\tilde{s}_i : i \in \mathcal{H}\}$ nor affects the seed-path
 837 computation, and is therefore orthogonal to both invariants.

838 A.4 Proof of Proposition 4 (Correctness of Algorithm 4)

839 Let $m = |\mathcal{S}| = s - 1$. The process of recovering the secret components is modeled
 840 as a *generalized coupon-collector problem* with group drawings [12,1], where each
 841 query draws a batch of w coupons out of m distinct types.

Correctness of Extract. For each scheme, Definition 5 specifies **Extract** as a closed-form algebraic procedure: matrix inversion over $\mathbb{F}_q$ (MEDS), column extraction (LESS), or a group operation (CROSS). Each procedure is deterministic and correct given the dual revelation $(\tilde{s}_i, \text{resp}_i)$: the algebraic relationship between the response and the hidden seed is an identity (not an approximation), so **Extract** returns exactly $\mathbf{sk}' =$ the secret component indexed by $j = c_i$.

We work under the assumption of Proposition 4: $v^* = \text{root}$, so the subtree of v^* is the entire tree and $L = |\mathcal{H} \cap \text{leaves}(\text{root})| = |\mathcal{H}| = w$ exactly—a deterministic quantity, not a random variable. Define

$$p_{\min} = 1 - \left(\frac{s-2}{s-1} \right)^w = 1 - \left(1 - \frac{1}{s-1} \right)^w.$$

For any $r \geq 1$ uncollected secrets, the probability that at least one of the w hidden leaves reveals a new secret index is

$$p_r = 1 - \left(1 - \frac{r}{s-1} \right)^w \geq 1 - \left(1 - \frac{1}{s-1} \right)^w = p_{\min} > 0,$$

where the inequality holds because $1 - r/(s-1)$ is decreasing in r , so $r = 1$ gives the weakest bound. Hence

$$\Pr[\text{new}(q) \geq 1 \mid \text{Collected}] \geq p_{\min} > 0 \quad \text{for all } s \geq 2, w \geq 1.$$

Let T_j be the number of effective queries to go from $|\text{Collected}| = j - 1$ to $|\text{Collected}| = j$. Conditional on $|\text{Collected}| = j - 1$, each query succeeds with probability at least $p_{\min}$, so T_j is stochastically dominated by a $\text{Geom}(p_{\min})$ random variable, which is finite almost surely. Hence $Q = \sum_{j=1}^m T_j$ is finite almost surely, and Algorithm 4 terminates with probability 1.

Expected query count and tail bound. Let R_k be the number of uncollected secret indices after k effective queries, and let $r = R_{k-1} > 0$. Each of the w hidden leaves (recall $v^* = \text{root}$, so all w hidden leaves are exposed) independently draws a uniform index from $\{1, \dots, s-1\}$, so the probability that *none* of the w draws falls in the remaining set of size r is $((s-1-r)/(s-1))^w$. Hence the probability of revealing at least one new index is

$$p_r = 1 - \left(\frac{s-1-r}{s-1} \right)^w = 1 - \left(1 - \frac{r}{s-1} \right)^w. \quad (4)$$

Since $1 - r/(s-1)$ is decreasing in r , we have $p_r \geq p_{\min} = 1 - (1 - 1/(s-1))^w$ for all $r \geq 1$.

The expected total number of effective queries is

$$\mathbb{E}[Q] = \sum_{r=1}^m \frac{1}{p_r} \leq \frac{m}{p_{\min}} = \frac{s-1}{p_{\min}}, \quad (5)$$

where the inequality uses $p_r \geq p_{\min}$ for all $r \geq 1$.

Remark 9. The bound in (5) should be seen as an upper-bound, as it treats the recovery process as just learning one secret per time. In practice, a query may reveal multiple unseen indices simultaneously, especially during the early stages of the algorithm when r is large. As a consequence the actual expected number of queries can be smaller than $(s-1)/p_{\min}$.

For the tail bound, each $T_j \leq k_0$ with probability at least $1 - (1 - p_{\min})^{k_0}$ (geometric tail), so by a union bound over $m = s-1$ stages,

$$\Pr[Q > k] \leq (s-1) \cdot (1 - p_{\min})^{\lfloor k/(s-1) \rfloor}.$$

Setting $(s-1)(1 - p_{\min})^{\lfloor k/(s-1) \rfloor} \leq \delta$ and solving gives $k \leq \lceil \frac{s-1}{p_{\min}} \cdot \ln \frac{s-1}{\delta} \rceil$, establishing the bound in Proposition 4.

Output correctness. When the loop terminates, $|\text{Collected}| = m$, and for each $j \in \{1, \dots, s-1\}$ there is exactly one entry $(j, \text{sk}'_j) \in \text{Collected}$ with $\text{sk}'_j = A_j^{-1}$ (or the corresponding secret for each scheme). The Collect step assembles these into $\hat{\text{sk}}$, which equals sk by correctness of Extract.

A.5 Derivation of Proposition 5 (Expected query count)

Setup. We work in the root-fault setting of Proposition 5: $v^* = \text{root}$, so $\ell = t$ and $L = w$ exactly (deterministic). By the Fiat-Shamir uniformity assumption, the challenge index c_i is uniform in $\{1, \dots, s-1\}$ independently for each $i \in \mathcal{H}$.

Distinct secrets per query. Conditioning on $L = w$, the number of *distinct* secret indices revealed in one effective query has expectation

$$\mathbb{E}[\#\text{distinct} \mid L = w] = (s-1) \left(1 - \left(1 - \frac{1}{s-1} \right)^w \right) = E_{\text{dist}}, \quad (6)$$

by linearity of expectation and symmetry of the uniform distribution.

Remark 10 (General fault location). When $v^* \neq \text{root}$ and L is a hypergeometric random variable with mean $\bar{w} = w\ell/t$, taking expectation over L gives

$$\mathbb{E}[\#\text{distinct}] = (s-1) \left(1 - \mathbb{E} \left[\left(1 - \frac{1}{s-1} \right)^L \right] \right).$$

Since $f(x) = \alpha^x$ with $\alpha = 1 - 1/(s-1) \in (0, 1)$ is convex, Jensen's inequality yields $\mathbb{E}[f(L)] \geq f(\bar{w}) = \alpha^{\bar{w}}$; hence

$$\mathbb{E}[\#\text{distinct}] \leq (s-1) \left(1 - \left(1 - \frac{1}{s-1} \right)^{\bar{w}} \right) =: E_{\text{dist}}^{\bar{w}}. \quad (7)$$

$E_{\text{dist}}^{\bar{w}}$ is an *upper* bound on the expected per-query gain, which implies a *lower* bound on $\mathbb{E}[Q]$. It cannot be used to derive an upper bound on $\mathbb{E}[Q]$; the upper bound in Proposition 5 follows solely from the direct inequality $p_r \geq p_{\min}$ established in the root-fault setting.

898 *Total queries (root fault).* From equation (4) (see page 29), $p_r = 1 - (1 - r/(s -$
899 $1))^w$ (with $L = w$ deterministic), and since $p_r \geq p_{\min}$ for all $r \geq 1$,

$$\mathbb{E}[Q] = \sum_{r=1}^m \frac{1}{p_r} \leq \frac{m}{p_{\min}} = \frac{s-1}{p_{\min}},$$

900 which for $p_{\min} \approx 1$ (all MEDS parameter sets, cf. Table 2) gives $\mathbb{E}[Q] \lesssim s - 1$.